1 The nature of technology: Understanding networks

You will learn

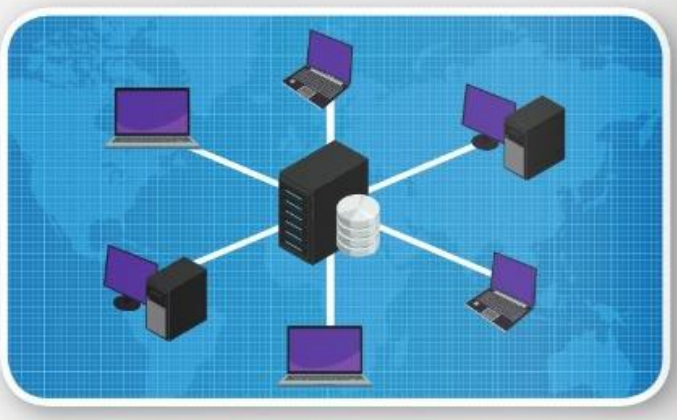
- about network hardware and how the components work together
- how messages are sent over the internet
- about packet switching
- how to connect to a network and how to solve problems with network connections
- about cloud storage and other services in the cloud.

In Student Book 5 you learned how to look for clues that tell you there is a network in your school. You learned that networks use special hardware such as hubs and routers. In this unit you will learn more about networks. You will learn about the hardware components of a network and how they are connected. You will learn how messages are sent over the internet. You will also learn how to connect to a network and how to solve some of the problems you might experience along the way. Finally, you will learn about cloud storage and other services in the cloud.



Unplugged

What do you know about network connections? Look around the room you are in and make a note of everything you can see that tells you there is a network in the room. If there is time, take pictures or make drawings.



Learning outcomes: Explain how computers communicate; Describe internet services (for example, cloud storage)

network hub
switch server router packets
packet switching cloud
data centre wireless access point



Talk about...

In 2019, more than half (around 56%) of the world's population had access to the internet.

The number of people with access to the internet varies in different parts of the world. The rates are high in Europe, North America and parts of Asia. In parts of Africa, access to the internet is still very low.

Is it fair that some people are unable to access the internet? What do you think people miss out on? Is there anything the rest of the world could do to make the internet easier to access?

Did you know?

A new improved mobile phone service called 5G was introduced in 2019. 5G provides an internet connection that is faster than home internet links that run over cables. Some people think that 5G phone networks will become the way that we access the internet at home in the future.

1.1 Network hardware

In this lesson

You will learn:

- about the hardware needed in a network.

Spiral back

In Student Book 5 you learned about two types of network. There are local area networks, like the one in your school, and wide area networks like the internet. Networks use special equipment. You learned how to look for clues that tell you there is a network in your school.

Using networks

When you use a computer, you are usually using a **network**.

You may use your school network. If you are at home, you may have a **broadband network** that connects you to the internet.

When you are sending an email, saving a school assignment or posting your thoughts on a social media site, you use a network. You often don't think about using a network. You save a document and expect it to be available next time you need it. But what happens to the file and where does it go?

When we send or receive a message over a network, we use many pieces of computer hardware. In this lesson you will learn more about network hardware.

Network hardware

Servers and storage

One of the most important pieces of hardware in a network is a **server**. A server is a computer.

The clue to what a server does is in the name. When you go to a restaurant, a waiter serves you food. The job of a waiter is to make sure everyone gets the meal they ordered.



A server stores and sends computer files and messages to users and to devices like printers. The job of a server is to make sure files go to the person or device that they are intended for.

Here are some examples of what a server can do.

It can:

- receive a word-processed file and save it to a network storage drive
- find a spreadsheet file you ask for and send it to your computer
- receive an email from you and pass it on to someone else
- receive a request from you to print a file and send the file to a printer.

Different types of servers do different jobs. For example, there are file servers, print servers and email servers.

Activity

What type of server is used for each of the examples?

What job do you think a web server does?

All computers need to store data. Servers are no different. They have storage devices attached to them. The storage devices used by servers are larger and faster than the ones in personal computers. This is because **network storage** is shared by many people who use the network.

Hubs and switches

When messages are sent out across a network, they need to be sent to the correct place. Hubs and switches do this job. They work in different ways.

When a **hub** receives a message, it sends it to all the computers that are connected to it. Every computer checks the message. The computer that the message is meant for keeps the message. The other computers ignore the message.

When a **switch** receives a message, it decides which computer the message is meant for. It sends the message only to that computer.

So, for example, if a hub is connected to 20 devices, it sends a message to all 20 devices. A switch only sends the message to the device it is meant for.



Hubs and switches: advantages and disadvantages

When a network is busy, the extra messages a hub sends out can slow the network down. A hub is also less secure than a switch. A hacker has more chance of stealing messages.

A switch is more expensive than a hub and is more difficult to set up. But it is more secure and works faster.

Activity

Imagine you work in a computer department. Your manager sends you this email.

I am planning to upgrade the network in our accounts department. The computer manager wants to use a switch for the network. I have been told a hub would be cheaper. What is your advice?

Write a short reply to the email.

Routers and modems



A **router** joins two networks together. Routers are usually used to connect a LAN or a home network to the internet.

When two networks are joined together, they sometimes use different ways to send data. It is like two people talking in different languages. They can communicate if a translator works with them. A translator understands both languages. A translator changes words from one language to the other. This is the job of a **modem** in a network. It takes data sent by one network and translates it into a form that can be understood by the second network.

A wireless access point



Most modern networks allow wireless connection. This means that devices like laptops and tablets can connect to the network without having to use a cable. A wireless connection needs a special piece of hardware called a **wireless access point** (WAP).

A WAP is connected to the network by a cable. But computers can connect to the WAP wirelessly, without a cable. Several computers can connect to a single WAP. WAPs are put on ceilings or high on walls.

A network interface card

A **network interface card** (NIC) allows a computer or other device to connect to a network. Most devices come with a NIC fitted.



A standard desktop or laptop computer will be fitted with two NICs.

- One allows a network cable to be plugged in.
- The other allows a wireless connection.

NICs are fitted inside the computer case but you will see the network cable socket on the outside of the case.



Activity

Look for network hardware at school. Take photos of the hardware you find.

Make a document and label the photos.

If you have a home network, take a photo of your home router and add it to the document.

Extra challenge

Search the web for images of the types of network hardware described in this lesson.

Paste the images at the end of the document you created in the activity. Label them.

Test

1. What does a server do in a network?
2. What piece of network hardware is used to give a wireless connection?
3. How do a router and modem work together in a network?
4. Explain why you would use a switch in a network instead of a hub.

1.2 How a network fits together

In this lesson

You will learn:

- how the hardware components of a network are connected
- how the hardware components work together.

Home network

Many people have internet connections at home. An internet signal reaches your home through telephone or special broadband cables. When the internet connection reaches your home, it is connected to a device called a home router.

In Lesson 1.1 you learned about the devices used in a network. A home router combines the main devices needed for a network into a single small case that sits on a shelf or desk. The small home router contains four main components.

- A **router** connects your home to the internet.
- A **modem** converts the signals that pass along telephone or broadband cables into digital data your home network can use.
- A **switch** makes sure that messages coming into your home are sent to the correct computer.
- A **wireless access point** lets you connect to the network wirelessly, from anywhere in your home.



The photo shows the back of a home router. The yellow cable links the internet to the router. The four empty sockets allow devices to be plugged into the router using a cable. The sockets are sometimes used to connect other devices like a printer.

The three aerials (which look like black sticks in this photo) are connected to the wireless access point. Not all WAPs have aerials, but aerials can improve the wireless signal. Most people connect computers to the router using a wireless connection.

Activity

Which components of a network listed in Lesson 1.1 are not included in a home router? List them.

Explore more

If you have a home broadband connection, examine the home router carefully. Can you identify the features described in this lesson?

Don't touch or remove any cables. It is an electrical device so can be dangerous.

Local area network

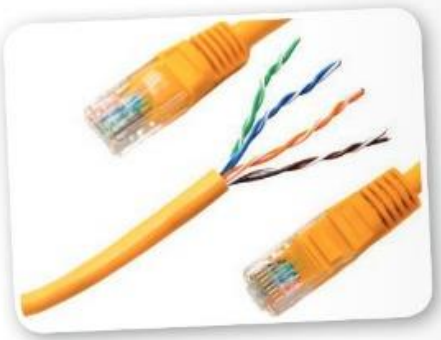
A **local area network** (LAN) is much bigger than a home network, but it contains the same components. The components in a LAN are larger and there are many more of them. A LAN is not stored in a single case like a home network. LAN equipment is spread around a building. The equipment is stored in cabinets that either stand on the floor or are attached to walls.

Connecting LAN components

Network components like servers and switches are connected by cables.

Two types of cable connect network equipment.

- **Copper cable** is the most common cable in a network. Networks use **twisted pair cables**, made up of pairs of thin copper wires twisted together. Data is sent along copper cables as pulses of electricity.
- **Fibre-optic cable** is made up of thin strands of clear fibre. Data is sent along **fibre-optic cables** as pulses of light.





There are three important differences in the two types of cable.

1. More data can be sent at faster speed along fibre-optic cables. Data is sent as light. Nothing travels faster than light.
2. Fibre-optic cables can be used over longer distances than copper. Electrical pulses fade as they travel. Copper cables can only be used over distances of 100 metres.
3. It is cheaper to use copper cable.

The server room

At the centre of a network is a **server room**. The room is usually air conditioned. This is because computer equipment makes a lot of heat. If it gets too hot, the equipment may break down.

The server room contains all the servers in the network. It also contains hubs and switches so that the rest of the network can be connected to the servers.

All servers must be connected to a hub or switch. Data passes along a single cable between the server and a hub or switch. All devices must be connected to a hub or switch. A single hub or switch can connect to many devices. Each connection uses its own cable.

When a computer is plugged into a network socket, it is connected to the network. It sends messages along a cable to the hub or switch. The hub or switch passes the messages from many devices to and from a server.



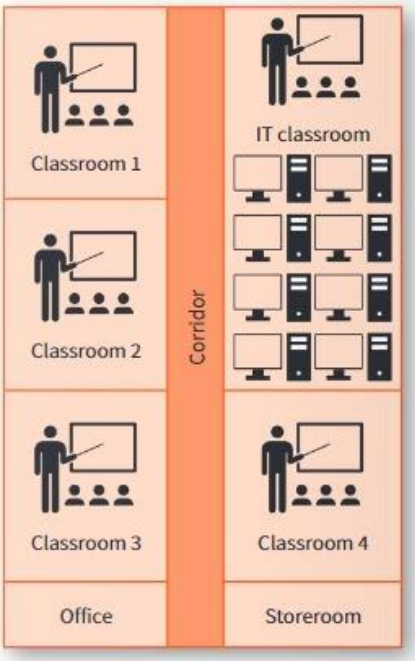
The hub room

Sometimes you need to connect an area of a building to a network that is more than 100 metres away from the server room. This can't be done with copper cable. The electrical signal won't transmit that far. So a **hub room** is set up close to the area that is going to be connected to the network.

A hub room does not contain servers. It only has switches and hubs. Every network socket in the distant area is connected to a hub or switch in the hub room. The hubs or switches in the hub room are connected to the main server room using a fibre-optic cable. Using a fibre-optic cable means that the network can be more than 100 metres from the server room.

Activity

The diagram shows an area in a school that is going to be connected to the school LAN. The area contains four classrooms, an IT classroom containing 20 computers, an office and a storeroom.



Each of the classrooms (including the IT classroom) must have:

- two network connections on the teacher's desk
- a network connection for a WAP installed on the ceiling.

The IT classroom should also have:

- a single network connection for each of the 20 computers
- a network connection for a printer.

The office will be used by four teachers. Each will have their own desk.

- Each desk will have two network connections.
- There will be an extra network connection for a printer in the office.

Draw a copy of the plan. In each room, write the number of network connections needed.

How many network connections are needed in total?

Extra challenge

Hubs or switches for the network will be kept in a cabinet on the wall of one of the rooms. Suggest two rooms that the cabinet could go in.

The school has two spare switches it can use. Each switch has 24 input connections. Does the school need to buy any more switches?

Test

1 List four network components that are contained in a home router.
2 What are the two types of cable used in LANs?
3 Give an example of where a fibre-optic cable is used in a LAN.
4 Explain why hub rooms are used in LANs.